

Math Hard Question 9

1

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ABC

Elizabeth has \$4500 in an account. Each year she expects to have **11%** more money in the account than the previous year. Which of the following models best describes how Elizabeth expects the money in her account to change over time?

A. Decreasing exponential

B. Decreasing linear

C. Increasing exponential

D. Increasing linear

2

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ABC

The function

$$T(x) = \frac{20,000 - 6.5x}{1,000}$$

gives the estimated air temperature $T(x)$, in degrees Celsius ($^{\circ}\text{C}$), at an altitude of x meters.

If the estimated temperature surrounding the hot air balloon is **15 $^{\circ}\text{C}$** , which of the following is closest to the altitude x of the hot air balloon?

A. 20

B. 769

C. 1,333

D. 5,385

3

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ABC

The function f is defined as:

$$f(x) = (x - a)(x - b)$$

where a and b are integer constants.

Given that: $f(17) > 0$, $f(20) < 0$, $f(23) > 0$

What is one possible value of $a + b$?

A. 39

B. 38

C. 37

D. 36

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ABC

The positive number a is 2,106% of the sum of the positive numbers b and c , and b is 81% of c .

What percent of b is a ?

A. 4,706%

B. 47.06%

C. 470600%

D. 47060%

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ABC

For the function f , for every increase of 2 in the value of x , the value of $f(x)$ increases by a factor of c , where c is a constant.

Which of the following equivalent forms of function f displays the value of c as the base or the coefficient?

A. $f(x) = 42(2)^6$

B. $f(x) = 42(8)^2$

C. $f(x) = 42(64)^x$

D. $f(x) = 42(4096)^{\frac{x}{2}}$

6

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ABC

A study investigated the effect of magnesium supplementation on sleep quality. 190 adults were randomly selected, and their sleep efficiency was measured.

Participants were randomly assigned to take a magnesium supplement or a placebo for 6 weeks. The study concluded that magnesium improved sleep quality.

What feature of this study supports this conclusion?

A. There were more than 100 participants in the study.

B. The participants were selected at random from the community center.

C. The sleep efficiency of each participant was measured again at the end of 6 weeks.

D. Each participant was randomly assigned to take a magnesium supplement or a placebo.

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ABC

A circle in the xy -plane has its center at $(-1, 1)$. Line t is tangent to this circle at the point $(7, -6)$.

Which of the following points also lies on line t ?

A. $(0, 7)$

B. $(6, 9)$

C. $(14, 2)$

D. $(15, 1)$

8

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ABC

The function $f(x)$ is defined by:

$$f(x) = \frac{x^2 + ax + b}{2x + c}$$

where a , b , and c are constants.

The graph of $f(x)$ does not intersect the line $x = 4$. If $f(5) = f(6) = 0$, what is the value of $a + b + c$?

A. 10

B. 11

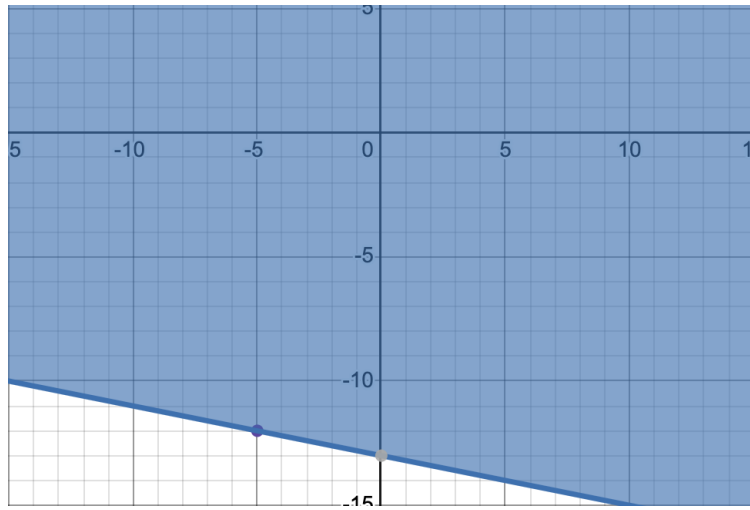
C. 12

D. 13

9

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ABC



The shaded region represents the solutions to the inequality:

$$rx + ty \geq -65$$

What is the value of $r + t$?

A. 6

B. 4

C. -4

D. -5